

Regina v. Terceira*

[Indexed as: R. v. Terceira]

38 O.R. (3d) 175
[1998] O.J. No. 428
Docket No. C15459

Court of Appeal for Ontario
Brooke, Finlayson and McKinlay JJ.A.
February 9, 1998

*Note: An appeal from the following judgment of the Ontario Court of Appeal to the Supreme Court of Canada (McLachlin C.J., Gonthier, Iacobucci, Major, Bastarache, Binnie and Arbour JJ.) was dismissed on December 16, 1999. See 46 O.R. (3d) 96 and [1999] S.C.J. No. 74.

Criminal law -- Evidence -- Admissibility of novel scientific evidence -- DNA evidence -- Trial judge holding voir dire on admissibility of novel science required to make threshold finding of reliability -- Threshold reliability to be decided on balance of probabilities -- Ultimate reliability issue for jury -- Trial judge not erring by limiting ruling on voir dire to issues raised by parties -- Appeal dismissed.

Criminal law -- Evidence -- Admissibility of DNA statistical evidence -- Accused seeking bar to admission of population frequency statistics regarding likelihood of random match between DNA profiles -- Experts providing differing statistical range -- Trial judge repeatedly warning jury not to be overwhelmed by magnitude of numbers -- Trial judge not erring in admitting statistical evidence -- Admissibility of statistics to be determined by exercise of discretion in each case -- Appeal dismissed.

Criminal law -- Evidence -- Prior consistent statements
-- Trial judge did not err in excluding prior exculpatory statement made by accused at time of his arrest.

Evidence -- Admissibility -- Hypnotically refreshed memory
-- Accused arguing on appeal that evidence of identification witness should not have been admitted as witness's memory hypnotically refreshed -- At trial accused attacking witness's credibility on basis evidence tainted by police giving witness information not due to hypnosis -- Evidence admissible
-- Hypnosis relevant to weight of evidence and question for jury -- Appeal dismissed.

The accused was charged with first degree murder. The body of the six-year-old victim was discovered in the boiler room of the apartment where she lived and where the accused worked as a janitor. The Crown alleged that the accused lured the victim to the boiler room, sexually assaulted her and killed her. Hair, fibre, blood and DNA evidence which matched the accused was left on the floor at the attack site and on the victim's clothing. DNA tests were conducted on semen found on the floor outside the boiler room and on the victim's clothing. DNA testing indicated that the victim's blood was mixed with the accused's semen.

It was agreed that asphyxia was the cause of the victim's death. The Crown's theory was that the accused had caused the asphyxia by placing his hands over the victim's mouth. The defence position was that the victim had suffered an epileptic seizure triggered in part by a pre-existing head injury, and that it may have been the seizure which caused the asphyxia. At trial, the Crown called the pathologist who performed the autopsy on the victim. He testified that he had been informed at the preliminary inquiry of a head injury that the victim had suffered six weeks before her death. It was his opinion that the skull injury which he observed at the autopsy was a post-mortem development. In cross-examination, he agreed that certain head injuries have the potential to initiate epilepsy and that an epileptic seizure can cause asphyxia. The defence called a forensic pathologist who testified that it was his opinion that the skull injury noted at the autopsy resulted

from a pre-existing injury to the victim's head which could have triggered an epileptic seizure which in turn could have caused the asphyxia. The trial judge permitted the Crown to call another expert in reply, who testified that the head injury was a post-mortem artifact and that even if the victim had suffered a head injury six weeks prior to her death, her death was unrelated to it.

The accused was convicted of first degree murder. He appealed. The main arguments on the appeal related to the admissibility of the DNA evidence and the instruction to the jury with respect thereto; the admissibility of a prior consistent statement given to the police by the accused when he was arrested; the admissibility of what the defence referred to as "hypnotically refreshed testimony" of a Crown witness; the propriety of permitting the Crown to call the reply evidence; and the trial judge's instruction to the jury on consciousness of guilt.

Held, the appeal should be dismissed.

The Supreme Court of Canada did not, in *R. v. Mohan*, introduce, as a precondition to admissibility, a new standard of proof in the scrutiny of opinion evidence that relates to a novel scientific theory or technique. The defence argued that, since DNA evidence is so compelling that it is dispositive of the guilt or innocence of the accused, there should be a burden on the Crown to satisfy the trial judge beyond a reasonable doubt that its proffered evidence was reliable, and that, in giving testimony on the voir dire, an expert could not rely upon opinions and tests supplied by others that are within the field of his or her expertise. To accede to the defence argument would be to give to DNA evidence a special status not generally accorded to the opinions of experts in criminal trials. The screening process engaged in by the trial judge is directed to the issue of admissibility of the novel expert testimony, not the determination of its ultimate reliability, which is a matter for the jury. The trial judge is merely required to determine on a balance of probabilities whether the evidence meets a threshold of reliability. His function is limited to an overview of the evidence proffered in order to be

satisfied that it reflects a scientific theory or technique that has either gained acceptance in the scientific community, or if not accepted, is considered otherwise reliable in accordance with the methodology validating it. The trial judge must not pass judgment on the particular application of the methodology by the expert. That is a question of weight to be determined by the jury. The nature and scope of the evidence necessary for the trial judge to reach the threshold determination will vary according to the type of evidence proffered and the concessions made by counsel. As a result, it would be unwise to define the threshold test of reliability with the precision advanced by the scientific community. Rather, the threshold test of reliability must remain capable of adaptation to changing circumstances and realities. Simply stated, the threshold test of reliability is met when the trial judge, having reviewed certain evidence presented by counsel, feels that the novel scientific technique or theory is sufficiently reliable to be put to the jury for its review.

At the conclusion of the evidence, the trial judge should advise the jury in the normal way as to the limits of the expert evidence and the use to which it can be put. Additionally, in the case of DNA evidence, he or she would be well advised to instruct the jury not to be overwhelmed by the aura of scientific infallibility associated with scientific evidence. The trial judge should tell them to use their common sense in their assessment of all of the evidence on the DNA issue and determine if it is reliable and valid as a piece of circumstantial evidence.

The accused argued that there should be a complete prohibition against the admission of frequency population statistics. The first step in the DNA process is concluded when an expert determines that there is a match (both visually and based on calculating the fragment length of the DNA profile) between the accused's DNA profile and an evidentiary sample, such as the semen located on the leotards. To appreciate the significance of the match, the next step is to express the likelihood of the match being random. The less frequency with which one observes a given DNA profile within the human population in general, the stronger the inference that the

evidentiary sample may have come from the accused. The accused argued that the frequency population statistics should be expressed only in qualitative terms such as "rare", "common" or "very rare". Crown and defence experts applied different methods of calculating the frequency of occurrence of the match resulting in widely varying statistics from 1 in 1,500 to 1 in 1.8 million with respect to one of the matches. The trial judge ruled that numerical expression of the significance of the match was appropriate in this case and that the jury would not be overwhelmed by numbers. The trial judge repeatedly warned the jury in his charge not to be overwhelmed or dazzled by the numbers when the evidence was adduced.

The trial judge did not err in admitting the statistics in this case. Qualitative descriptors do not accurately convey the potency of a match and are imprecise. DNA evidence is a form of identification, differing primarily from other forms of identification evidence in its ability to discriminate among individuals and to exclude innocent persons as suspects. A concern raised by defence counsel is that the jury may misuse the evidence by viewing it as the expression of the likelihood of an accused's innocence, rather than as a measure of the likelihood of a random match between the accused's DNA profile with DNA profile of the evidentiary sample. This potential misuse of the evidence is known as the "prosecutor's fallacy". However, defence counsel conceded that there was no prosecutor's fallacy in this case as both the experts giving the evidence, and the charge to the jury, clearly warned the jury against this potential pitfall. There is no basis on the record in this case to support an argument that the jury would have misused the statistics as a predictor of the likelihood of guilt.

The trial judge refused to admit an exculpatory statement made by the accused to the police at the time of his arrest, and also refused to allow the defence to inform the jury of the fact that the accused was questioned when he was arrested and that he provided a statement. He ruled that the accused's denial of guilt when he was arrested was but one of a number of exculpatory statements made after the accused was aware of his status as a suspect, and was so similar in substance to earlier

statements that no new or helpful information would be provided by admitting the statement. That ruling was in line with the traditional Canadian position that defence counsel is not entitled to elicit an accused's prior exculpatory statements since prior consistent statements are irrelevant and superfluous. The trial judge did not err in excluding the prior exculpatory statement.

A Crown witness had given statements to the police contradicting the accused's alibi defence. However, her statements suggested that she was recalling the events of another day. The police arranged for her to see a psychologist in order for her to be hypnotized as a way of refreshing her memory. Before she saw the psychologist, the police advised her of "errors" in her memory of the events of the day that the victim disappeared. Following her hypnosis, the witness gave a statement to the police which was consistent with the testimony she provided at trial. While the defence characterized her evidence as "hypnotically refreshed testimony", no objection was actually taken at trial to the hypnosis itself, and trial counsel did not challenge the validity of hypnosis as a reliable scientific technique. Rather, the primary focus of the objection to the admissibility of the evidence was on the tainting of the evidence by the police. Defence counsel at trial conducted an effective cross-examination of the witness in which he pointed out inconsistencies in her statements, impeached her credibility through use of the transcript of her hypnosis session, and highlighted deficiencies in her recollection which were unaffected by hypnosis. During the testimony of the hypnotist and the police officer who interviewed the witness, defence counsel brought out the areas in which the police had "corrected" the witness's recollection. The trial judge gave an instruction to the jury as to the special caution that should be exercised in assessing her evidence due to the fact that she had been hypnotized. There was no reason to give effect to this ground of appeal.

The trial judge did not err in permitting the Crown to call the expert reply evidence. The Crown was not splitting its case. The fact that the pathologist who conducted the autopsy had responded to certain hypothetical questions on the premise

that there was a pre-existing fracture did not place an onus on the Crown to chase down and eliminate this possibility that the asphyxia had resulted from a fracture which was not evident to the pathologist. It was not until the defence expert put this theory forward affirmatively that it took on real significance and presented the jury with an alternative cause of death which could have impacted on the verdict. The Crown did not change its position; it maintained it by reacting to a defence witness who suggested a different cause of death.

The trial judge did not err in failing to direct the jury to apply the standard of proof beyond a reasonable doubt to the evidence of consciousness of guilt. Evidence of consciousness of guilt should not be considered in isolation, and should not have the standard of proof beyond a reasonable doubt applied to it separately from the rest of the evidence.

R. v. Campbell (1978), 17 O.R. (2d) 673, 38 C.C.C. (2d) 6, 1 C.R. (3d) 309 (C.A.); R. v. White (1996), 29 O.R. (3d) 577, 108 C.C.C. (3d) 1, 49 C.R. (4th) 97 (C.A.), apld

R. v. Biddle, [1995] 1 S.C.R. 761, 22 O.R. (3d) 128n, 123 D.L.R. (4th) 22, 178 N.R. 208, 96 C.C.C. (3d) 321, 36 C.R. (4th) 321; R. v. Storey (1968), 52 Cr. App. R. 334 (C.A.), distd

R. v. Mohan, [1994] 2 S.C.R. 9, 29 C.R. (4th) 243, 89 C.C.C. (3d) 402, consd

Other cases referred to

Frye v. United States, 293 F.1013; R. v. Abbey, [1982] 2 S.C.R. 24, 39 B.C.L.R. 201, 138 D.L.R. (3d) 202, 43 N.R. 30, [1983] 1 W.W.R. 251, 68 C.C.C. (2d) 394, 29 C.R. (3d) 193; R. v. B. (K.G.), [1993] 1 S.C.R. 740, 148 N.R. 241, 79 C.C.C. (3d) 257, 19 C.R. (4th) 1; R. v. Beland, [1987] 2 S.C.R. 398, 43 D.L.R. (4th) 641, 79 N.R. 263, 36 C.C.C. (3d) 481, 60 C.R. (3d) 1; R. v. Corbett, [1988] 1 S.C.R. 670, 28 B.C.L.R. (2d) 145, 85 N.R. 81, [1988] 4 W.W.R. 481, 34 C.R.R. 54, 41 C.C.C. (3d) 385, 64 C.R. (3d) 1; R. v. Court (1995), 23 O.R.

(3d) 321, 99 C.C.C. (3d) 237, 29 C.R.R. (2d) D-1 (C.A.); R. v. Egger, [1993] 2 S.C.R. 451, 103 D.L.R. (4th) 678, 153 N.R. 272, 15 C.R.R. (2d) 193, 82 C.C.C. (3d) 193, 21 C.R. (4th) 186, 45 M.V.R. (2d) 161; R. v. Johnston (1992), 69 C.C.C. (3d) 395, 12 C.R. (4th) 99 (Ont. Gen. Div.); R. v. Lavallee, [1990] 1 S.C.R. 852, 67 Man. R. (2d) 1, 108 N.R. 321, [1990] 4 W.W.R. 1, 55 C.C.C. (3d) 97, 76 C.R. (3d) 329; R. v. Melaragni (1992), 73 C.C.C. (3d) 348 (Ont. Gen. Div.); R. v. Melnichuk, [1997] 1 S.C.R. 602, 146 D.L.R. (4th) 686, 209 N.R. 321, 114 C.C.C. (3d) 503, revg (1996), 104 C.C.C. (3d) 160 (Ont. C.A.); R. v. Morin, [1988] 2 S.C.R. 345, 30 O.A.C. 81, 88 N.R. 161, 44 C.C.C. (3d) 193, 66 C.R. (3d) 1; R. v. Parsons, [1980] 1 S.C.R. 785, 110 D.L.R. (3d) 71, 33 N.R. 158, 51 C.C.C. (2d) 350, 14 C.R. (3d) 191, affg (1977), 17 O.R. (2d) 465, 80 D.L.R. (3d) 430, 33 N.R. 161, 37 C.C.C. (2d) 497 sub nom. Charette v. R. (C.A.); R. v. Scardino (1991), 46 O.A.C. 209, 6 C.R. (4th) 146 (C.A.); R. v. Small, unreported, September 11, 1991 (Ont. Gen. Div.); R. v. U. (F.J.), [1995] 3 S.C.R. 764, 128 D.L.R. (4th) 121, 186 N.R. 365, 101 C.C.C. (3d) 97, 42 C.R. (4th) 133

Statutes referred to

Criminal Code, R.S.C. 1985, c. C-46

Authorities referred to

Matthews, Pink, Tupper and Wells, The Expert, A Practitioners Guide (Toronto: Carswell, 1995), vol. 1, ch. 12, p. 12-1
 Sopinka, Lederman and Bryant, The Law of Evidence in Canada (Toronto: Butterworths, 1992), p. 359

APPEAL by the accused from his conviction for first degree murder.

Russell S. Silverstein and David M. Tanovich, for appellant.
 Carol A.R. Brewer, Michal Fairburn and Shawn D. Porter, for the Crown, respondent.

The judgment of the court was delivered by

FINLAYSON J.A.: -- The appellant was tried in the Ontario Court (General Division) before the Honourable Mr. Justice A. Campbell and a jury on a plea of not guilty to an indictment charging that the appellant:

. . . on or about the 14th day of October in the year 1990, at the Municipality of Metropolitan Toronto, in the Toronto Region did kill Andrea Atkinson and thereby commit First Degree Murder, contrary to the Criminal Code.

On February 4, 1993, after 56 days of trial and three days of jury deliberation, the appellant was convicted of first degree murder. He was sentenced to life imprisonment with minimum parole eligibility after 25 years.

OVERVIEW OF THE FACTS

On October 14, 1990, shortly after 9:00 a.m., Andrea Atkinson said goodbye to her mother, Ruth Windebank and left her apartment to play outside. She was last seen outside the apartment building at 9:30 a.m. by Rosemaria Lorengard. Around 11:00 a.m., Andrea's mother began looking for her. Over the next two hours, she checked with neighbours, with Andrea's friends and around the apartment building. She found no sign of her daughter. Sometime after 1:30 p.m., Ruth Windebank called 911 and reported that her daughter had gone missing.

During the next nine days, an extensive search was conducted for Andrea. On October 23, 1990, her body was discovered by accident by Elese Roberts, a janitor, and John Clarke, a maintenance supervisor, in the sixth floor boiler room of the apartment building where she and her mother lived. Forensic examinations established that she had been sexually assaulted (there was a tear and bruising to her vagina, and semen was found on her leotards). The cause of death was determined to be asphyxia.

The appellant worked as a janitor at the apartment building

where Andrea lived. He was working on October 14, 1990. Two residents of the apartment building, Frank Burkett and Corinna MacNaughton, saw him at the building after Andrea was last seen alive. The Crown alleged that the appellant saw Andrea in the area and lured her to the sixth floor where he sexually assaulted and smothered her. He then hid the body behind a hot water tank in the boiler room. He was charged with murder on December 3, 1990.

The appellant had demonstrated his interest in Andrea by speaking to her on occasion and by ruffling her hair with his hand. According to Andrea's mother, Andrea had liked the appellant and often talked about him. He was her hero. The appellant had once chased away some boys who were bothering Andrea and her friends. Michelle Martin, Andrea's best friend, agreed that Andrea "looked up" to the appellant following this intervention on her behalf.

Curiously, the location of the boiler room on the sixth floor where the rape and murder of the victim took place was not well known, even to occupants of the apartment building. This is most clearly evidenced by the fact that the police, despite a massive search effort, failed to discover that there was a sixth floor with such a room on it.

The appellant knew about the boiler room's existence. He had keys to it and admitted that he went there on occasion to smoke drugs. While other janitors in the building had keys, the appellant was the only one scheduled to work the day the victim went missing.

Forensic evidence linked the appellant to the murder in the following ways:

- (a) Hair, fibre, blood and DNA evidence, which matched the appellant, was left on the floor at the attack site and on the victim's clothing.
- (b) A mixed blood and semen stain was found on the concrete floor outside the boiler room at the base of the stairs. The semen found on the floor was sufficient in quantity to

conduct both conventional serology and DNA testing. Semen was also found on a great deal of the victim's clothing. A semen stain found in her leotards was of sufficient quality and quantity to conduct DNA testing.

- (c) DNA testing indicated that the victim's blood was mixed with the appellant's semen. The victim bled from a "severe" tear to her vaginal area, which was one by two by three centimetres in size. The tear extended almost to her anus. Dr. McAuliffe, who conducted the autopsy, found bruising to the membrane of the vagina, suggesting that the injury had occurred prior to death.
- (d) Numerous blue fibres consistent with blue fibres from the outer portion of the sweat pants that the appellant wore to work on October 14, 1990 were found on Andrea's clothing and on the floor outside the boiler room.

The Crown relied on the appellant's conduct following Andrea's disappearance as evidence of consciousness of guilt on his part. In some of his earlier pre-arrest statements to the police, the appellant had lied about the number of times he had been in the boiler room and in the area outside the boiler room:

- (a) On October 23, 1990, in a statement to Constables Aitchison and McPhearson, the appellant maintained that he had been in the boiler room area only once to help another janitor retrieve a football which was on the roof.
- (b) On October 30, 1990, in a statement to Detectives Gauthier and McNamara, the appellant maintained that he had only been in the boiler room area once. The appellant was then confronted with incriminating evidence and a suggestion that another janitor lent him his keys, whereupon he admitted that he had discovered Andrea Atkinson's body in the boiler room while smoking some drugs several days before her ultimate discovery by Roberts and Clarke.

The Crown alleged that the appellant had lied to the police on October 30, 1990 when he told them that he had "discovered"

Andrea's body a few days after her disappearance. The Crown suggested that this was a fabrication to explain away the fibre and forensic evidence when he was confronted with it.

It was the theory of the defence that the appellant had nothing to do with the sexual assault and murder of Andrea Atkinson. The defence rested on alibi. The appellant alleged that he attended work on the morning of October 14, 1990 but that he became ill and left by 9:30 a.m. He maintained that he arrived home at 11:00 a.m., went to bed, and stayed there until 4:00 p.m. He then received a telephone call from a friend and tenant at the apartment building who told him that Andrea had gone missing. The appellant's mother and grandmother testified at trial to support the alibi. Both of these witnesses, Maria Mendes and Maria Terceira, had their credibility undermined by prior inconsistent statements put to them in cross-examination. While the defence challenged the cause of death, suggesting that the victim may have died from the onset of an epileptic seizure triggered by a pre-existing "growing fracture" of the skull, the defence position remained unequivocal that the appellant was not present in the apartment building at the time of the killing and that he did not commit the murder.

ISSUES

There were a number of grounds for appeal which were abandoned and one, relating to the right to challenge for cause, was simply reserved pending the outcome of appeals on this issue presently before the Supreme Court of Canada. I propose to deal with the following issues to which the Crown was called upon to respond during the hearing of this appeal.

- (1) The admissibility of the DNA evidence and the instruction to the jury with respect thereto.
- (2) The admissibility of a prior consistent statement given to the police by the appellant.
- (3) The admissibility of hypnotically refreshed testimony.
- (4) The propriety of permitting the Crown to call Dr. Humphries

in reply.

(5) The instruction to the jury on consciousness of guilt.

(1) The DNA evidence

By way of summary, this issue involves the appellant's submission that the trial judge failed to properly determine, as a preliminary matter, the admissibility of the DNA evidence proffered by the Crown. Specifically, case-law has required that certain threshold determinations be made by a trial judge in the absence of the jury before certain evidence can be offered at trial for closer scrutiny.

I am grateful to Matthews, Pink, Tupper and Wells, the authors of *The Expert, A Practitioners Guide* (Toronto: Carswell, 1995) at vol. 1, c. 12, *Forensic DNA Typing Evidence*, p. 12-1 and following, for a readable overview of DNA evidence. The introduction is reassuring:

DNA (Deoxyribonucleic Acid) typing may have caught the legal community by surprise, but it is merely an extension of the rapid evolution of molecular biology that began gaining momentum over the last twenty years. In 1995, the science and its forensic application are anything but novel, although forensic scientists are continuously seeking new means to improve sensitivity of detection, increase efficiency, and concomitantly decrease time of analysis.

DNA profiling is a comparatively new method of providing identification evidence for use in criminal cases. DNA evidence is used essentially for two purposes. The first use of DNA evidence is as evidence that the suspect's DNA "matches" the DNA found in blood, semen or tissue recovered at a crime scene. In this way, the DNA evidence serves an exclusionary purpose. In the absence of further qualifications, a "match" is no more than a failure to exclude a suspect's DNA from the crime scene. The debate at trial with respect to the determination of a match, as was the case during the trial of this matter, will often focus on the methodology used to determine a match. The second branch of the analysis of DNA evidence involves the application of

population genetics. Probability statistics are introduced in an attempt to bolster the significance of a "match". The scientist determines, according to an established database of known DNA samples, the statistical likelihood that another individual person would have the same DNA pattern as that of the suspect. Simply stated, this second branch considers the statistical likelihood of a random DNA match. Cross-examination of the expert tendering DNA evidence serving this second purpose will usually focus on the methodology used to calculate the numbers reflecting the frequency of the DNA pattern. The DNA evidence in the present case was used by the Crown for the above two purposes.

The DNA testing in the case in appeal took place between November 1990 and May 1991, and in the submission of the appellant, DNA profiling was then a novel scientific technique. This trial took place prior to the decision of the Supreme Court of Canada in *R. v. Mohan*, [1994] 2 S.C.R. 9, 29 C.R. (4th) 243, and accordingly the trial judge did not have the benefit of the judgment of Sopinka J. calling for special scrutiny in dealing with novel scientific theory or technique. However, the trial judge did hold an extensive voir dire before admitting this evidence and on that hearing the defence challenged in detail the reliability and validity of the opinions offered by the Crown's experts. The appellant challenges the sufficiency of the trial judge's consideration as a result of the voir dire with respect to the admissibility of the DNA evidence.

Both Crown and defence counsel on the DNA voir dire devoted a considerable portion of their submissions to a discussion of the standard to be applied in relation to the admission of novel scientific evidence. Crown counsel discussed the standard of "relevancy and helpfulness" as well as "relevancy and reliability". Defence counsel made submissions in favour of the adoption of the more restrictive "Frye" test articulated by the United States Supreme Court in *Frye v. United States*, 293 F.1013. Both counsel explicitly referred the trial judge to the decision in *R. v. Johnston* (1992), 69 C.C.C. (3d) 395, 12 C.R. (4th) 99 (Ont. Gen. Div.), wherein Langdon J. adopted a "reliability" standard. The trial judge characterized the defence position on the voir dire as urging "that the Crown has

not produced sufficient evidence on the voir dire to support the reliability and admissibility of Pamela Newall's techniques in analysis". The foregoing demonstrates that the trial judge was aware that initial determinations of reliability would be required before the proposed DNA evidence could be proffered at trial. Moreover, the appellant concedes that the trial judge recognized that reliability was a preliminary finding of fact that would need to be made before the proposed DNA evidence was admissible.

As it turns out, the precedential value of the DNA testing conducted in the present case is limited because it is conceded by all counsel that whatever its strengths and weaknesses in 1990-91, the techniques employed in this case are no longer in use. Accordingly, since the DNA testing was unique to this case, I do not propose to deal with the particularity of it. In the view I take of this opinion evidence, it is only the judicial process that led to its admissibility and the instructions to the jury with respect to its use that I need to consider.

Relying upon the judgment of the Supreme Court of Canada in *R. v. Mohan*, supra, counsel for the appellant submits that before the jury can be permitted to hear the evidence of DNA testing, the trial judge is required as a matter of law to conduct what he calls a "Mohan type hearing" in order to satisfy himself beyond a reasonable doubt as to the reliability of the evidence adduced by the experts for the Crown. By this counsel for the appellant suggests that the trial judge must satisfy himself as to the acceptance of the technology in the scientific community, the expertise of the Crown witnesses in that field, and the accuracy of the tests carried out pursuant to that technology, among other factors. All this to the criminal standard of proof. Then, and only then, can the same evidence be recalled for the consideration of the jury.

I have some considerable difficulty with this submission which, with respect, reflects a misreading of *Mohan*, supra. In my opinion, the rules laid down by Sopinka J. in *R. v. Mohan*, supra, do not signify a departure from the common law rules relating to the admission of opinion evidence in a criminal

trial, nor do they purport to do so. The four criteria for the admissibility of expert testimony are derived from case-law. They are set out by Sopinka J. as follows at p. 20 S.C.R., p. 252 C.R.:

- (a) relevance;
- (b) necessity in assisting the trier of fact;
- (c) the absence of an exclusionary rule;
- (d) a properly qualified expert.

Prior to *Mohan*, when relevant expert opinion evidence has been proffered, Canadian courts focused on two factors in determining its admissibility: the special knowledge criterion and the expertise criterion. In *R. v. Abbey*, [1982] 2 S.C.R. 24 at p. 42, 68 C.C.C. (2d) 394 at p. 409, Dickson J. provided the following formulation of the "special knowledge" requirement for the admissibility of expert evidence:

With respect to matters calling for special knowledge, an expert in the field may draw inferences and state his opinion. An expert's function is precisely this: to provide the judge and jury with a ready-made inference which the judge and jury, due to the technical nature of the facts, are unable to formulate. "An expert's opinion is admissible to furnish the Court with scientific information which is likely to be outside the experience and knowledge of a judge or jury. If on the proven facts a judge or jury can form their own conclusions without help, then the opinion of the expert is unnecessary": (*R. v. Turner* (1974), 60 Cr. App. R. 80, at p. 83, per Lawton L.J.).

The judgment of McIntyre J. is to much the same effect in *R. v. Beland*, [1987] 2 S.C.R. 398 at p. 415, 36 C.C.C. (3d) 481 at pp. 493-94:

The function of the expert witness is to provide for the jury or other trier of fact an expert's opinion as to the significance of, or the inference which may be drawn from,

proved facts in a field in which the expert witness possesses special knowledge and experience going beyond that of the trier of fact. The expert witness is permitted to give such opinions for the assistance of the jury. Where the question is one which falls within the knowledge and experience of the triers of fact, there is no need for expert evidence and an opinion will not be received.

In *Mohan*, supra, Sopinka J. quoted the above passage by Dickson J. from *R. v. Abbey*, supra, and went on to discuss the criteria of necessity at pp. 23-25 S.C.R., p. 254 C.R.:

This precondition is often expressed in terms as to whether the evidence would be helpful to the trier of fact. The word "helpful" is not quite appropriate and sets too low a standard. However, I would not judge necessity by too strict a standard. What is required is that the opinion be necessary in the sense that it provide information "which is likely to be outside the experience and knowledge of a judge or jury": as quoted by Dickson J. in *R. v. Abbey*, supra. As stated by Dickson J., the evidence must be necessary to enable the trier of fact to appreciate the matters in issue due to their technical nature. In *Kelliher (Village) v. Smith*, [1931] S.C.R. 672, at p. 684, this court . . . stated that in order for expert evidence to be admissible, "[t]he subject matter of the inquiry must be such that ordinary people are unlikely to form a correct judgment about it, if unassisted by persons with special knowledge".

As in the case of relevance, the need for the evidence is assessed in light of its potential to distort the fact-finding process. . . . The possibility that evidence will overwhelm the jury and distract them from their task can often be offset by proper instructions.

There is also a concern inherent in the application of this criterion that experts not be permitted to usurp the functions of the trier of fact. Too liberal an approach could result in a trial's becoming nothing more than a contest of experts with the trier of fact acting as referee in deciding which expert to accept.

These concerns were the basis of the rule which excluded expert evidence in respect of the ultimate issue. Although the rule is no longer of general application, the concerns underlying it remain. In light of these concerns, the criteria of relevance and necessity are applied strictly, on occasion, to exclude expert evidence as to an ultimate issue. Expert evidence as to credibility or oath-helping has been excluded on this basis. See *R. v. Marquard*, [1993] 4 S.C.R. 223, per McLachlin J.

It is to be observed that the word "reliable" is not listed among Sopinka J.'s four criteria. It is, however, discussed under "relevance" under his "cost-benefit analysis" as to whether expert evidence that is otherwise logically relevant may be excluded on the basis that its probative value is overborne by its prejudicial effect. He says at p. 21 S.C.R., p. 252 C.R.:

While frequently considered as an aspect of legal relevance, the exclusion of logically relevant evidence on these grounds is more properly regarded as a general exclusionary rule (see *R. v. Morris*, [1983] 2 S.C.R. 190). Whether it is treated as an aspect of relevance or an exclusionary rule, the effect is the same. The reliability versus effect factor has special significance in assessing the admissibility of expert evidence.

There is a danger that expert evidence will be misused and will distort the fact-finding process. Dressed up in scientific language which the jury does not easily understand and submitted through a witness of impressive antecedents, this evidence is apt to be accepted by the jury as being virtually infallible and having more weight than it deserves.

(Emphasis added)

As an introduction to his conclusion with respect to novel scientific theory or technique. Sopinka J. (at pp. 21-22 S.C.R., pp. 252-53 C.R.) quotes with approval the language of Moldaver J. in *R. v. Melaragni* (1992), 73 C.C.C. (3d) 348 (Ont.

Gen. Div.). Moldaver J. had applied "a threshold test of reliability" to such novel evidence and asked himself the following questions, among others:

(1) Is the evidence likely to assist the jury in its fact-finding mission, or is it likely to confuse and confound the jury?

(2) Is the jury likely to be overwhelmed by the "mystic infallibility" of the evidence, or will the jury be able to keep an open mind and objectively assess the worth of the evidence?

Sopinka J. picks up on this phrase "threshold test of reliability" with respect to novel scientific theory or technique. The focus of attention in this court by counsel was on the following summary by Sopinka J. at p. 25 S.C.R., p. 255 C.R.:

In summary, therefore, it appears from the foregoing that expert evidence which advances a novel scientific theory or technique is subjected to special scrutiny to determine whether it meets a basic threshold of reliability and whether it is essential in the sense that the trier of fact will be unable to come to a satisfactory conclusion without the assistance of the expert. The closer the evidence approaches an opinion on an ultimate issue, the stricter the application of this principle.

This statement of the law is clearly one of general application, but much of what is later said by Sopinka J. is specific to Mohan itself. In that case, the impugned evidence was that of a psychiatrist proffered by the defence who was prepared to testify as to the psychosexual profiles of those persons likely to have committed the sexual assaults of which the accused, a physician, stood charged. The thrust of the testimony was that the perpetrator of the offences alleged to have been committed would be part of a limited and unusual group of individuals and that the accused did not fall within that narrow group. In beginning his analysis of the ruling of the trial judge rejecting the admission of this evidence,

Sopinka J. made reference to the competing exclusionary rules. He said at p. 20 S.C.R., p. 251 C.R.:

The admissibility of the rejected evidence was analyzed in argument under two exclusionary rules of evidence: (1) expert opinion evidence, and (2) character evidence. I have concluded that, on the basis of the principles relating to exceptions to the character evidence rule and under the principles governing the admissibility of expert evidence, the limitations on the use of this type of evidence require that the evidence in this case be excluded.

Accordingly, the contest in Mohan as to admissibility was between the introduction of the novel opinion of the expert and the prohibition against introducing evidence which is directed towards demonstrating that the accused person has or has not an abnormal propensity to commit the crime in issue. The trial judge had ruled that the proposed evidence was inadmissible as going beyond evidence of general reputation and did not fall within the proper sphere of expert testimony. It was against this background that Sopinka J. stated at p. 37 S.C.R., p. 264 C.R.:

Before an expert's opinion is admitted as evidence, the trial judge must be satisfied, as a matter of law, that either the perpetrator of the crime or the accused has distinctive behavioral characteristics such that a comparison of one with the other will be of material assistance in determining innocence or guilt.

In the appeal before this court, there is no conflict with the rules relating to character evidence as such. Rather, the tension is between the probative value of the opinion evidence versus its prejudicial effect in the sense that its effect on the jury may be out of proportion to its reliability. Mohan, supra, stands as authority for the proposition that expert evidence which may be logically probative of an issue at trial may be nonetheless excluded in certain circumstances. Additionally, in light of the judicial reasoning from Mohan, supra, since we are confronted with what was at the time of trial perceived to be a novel scientific theory or technique,

we are concerned with the threshold issue of reliability, i.e., is the science itself valid. As I understand Mohan, with reference to the case in appeal, the requirement of a basic threshold of reliability as a pre-condition to admissibility is met where the trial judge is satisfied as to the reliability of DNA profiling as a novel scientific technique. Where the Crown and defence part company is with respect to the extent of the inquiry necessary to establish this pre-condition.

Our task is considerably narrowed by the concession of appellant's counsel that he is not suggesting that DNA profiling has not been found reliable in other jurisdictions. The appellant does not take issue with the microbiological aspects of DNA profiling. No general concern was raised at trial about the ability of the Centre of Forensic Science ("CFS") to extract DNA from biological substances and to isolate and remove regions on human chromosomes which are suitable for testing nor to determine whether any two samples were a "match" one to the other. Nor is counsel for the appellant suggesting that the process used by the CFS in this case, involving RFLP or "restriction fragment length polymorphism" analysis, is not an accepted methodology for DNA profiling. Rather, the complaint was that the DNA laboratory was only established by the CFS a few months prior to the testing in this case and there was no general acceptance of its specific methodology used to determine the statistical likelihood of a random match. The attack was not upon the technology of DNA profiling per se but upon the ability of the CFS, notably its principal expert Pamella Newall, to reliably utilize it. In addition, the appellant challenged the introduction of the probability figures as their prejudicial effect would exceed the probative value of presenting quantitative statements of random match probability as opposed to qualitative measures.

The issue respecting admissibility is further narrowed by the decision of the defence at trial to withhold its evidence casting doubt upon the accuracy of the technology employed by the CFS until after the trial judge had ruled the Crown's evidence admissible. Accordingly, much of the argument in this court on the reliability of the Crown's evidence is based on

evidence that the trial judge did not hear on the voir dire. The focus of the attack on the ruling of the trial judge as to admissibility can be dealt with under three headings:

- (a) the sufficiency of Ms. Newall's credentials in this specific area as revealed from her cross-examination;
- (b) her reliance upon hearsay reports and the results of tests performed by others; and
- (c) the trial judge's failure to make findings as to her credibility and the weight to be given to her evidence.

I think I should state briefly my position on these issues and then develop my reasoning later. As to (a) above, counsel for the appellant took us through the attacks made during the voir dire on the scientific validity of the evidence of Ms. Newall and invited the court to re-assess the reliability of her evidence. However, it is not the function of an appellate court to retry these factual issues and substitute our findings for those of the trial judge or make findings where he has declined to do so. As to (b) above, there is abundant authority for the proposition that an expert can rely upon hearsay reports and tests that are within the scope of his or her expertise. Finally, as to (c) above, I do not think that it was appropriate for the trial judge to do anything more than he did in assessing the reliability of this expert testimony. Specifically, I do not agree that the trial judge was obliged to be satisfied beyond a reasonable doubt that the testimony of Ms. Newall with respect to the DNA testing by the CFS, as reflected in the form of her conclusions, was reliable. Given that the technology existed and was generally accepted in the scientific community, the contest as to the validity of its application to the particular case was in the last analysis a matter for the jury to assess. An important distinction must be drawn between assessing the reliability of a methodology and determining the propriety of the application of the methodology in particular factual circumstances. The latter determination is strictly for the jury, while the former threshold determination is the responsibility of the trial judge.

The issue comes down to what is encompassed in the requirement that the trial judge is to satisfy himself that the DNA evidence (accepting it as novel) meets a basic threshold of reliability. For myself, I think it is a mistake for this court to lay down a structure that must be adhered to in every case involving novel scientific theory or technique. We have seen too many trials unnecessarily delayed because of rigid formalism in the consideration of problems relating to the admissibility of evidence. I would prefer to leave it to the judgment of the trial judge as to how far he must go in meeting this threshold of reliability in the particular case. This said, trial judges confronted with evidence of novel scientific theory or technique may seek guidance in satisfying their threshold test of reliability in light of previous case-law. Specifically, Johnston, supra, and Melaragni, supra, list factors which, in addition to others which may arise on the particular facts of a case, are helpful measures of reliability. Whereas here, the issues have been narrowed by the very proper concessions of counsel, it is hardly necessary to listen to an extended presentation of the general acceptance of DNA profiling as a reliable technique for what has been called "genetic fingerprinting". I think that the trial judge focused on the issues as outlined by counsel. Specifically, the trial judge considered and scrutinized the reliability of the determination of the match and the calculation of the frequency of a random match. In doing so, he had to consider in overview the nature of the proposed evidence and its foundation in science. He had to consider whether Pamela Newall, the Crown's expert who was a research forensic biologist at the CFS, had the necessary expertise to enable her to express an opinion in this field. In this case, as I have stated, the existence of the technology itself was not in issue. The dispute was restricted to the specific nature and content of the expert evidence in this field as adduced through the testimony of Pamela Newall. In short, the trial judge was required in the case on appeal to make an inquiry as to whether Ms. Newall's evidence met a threshold of reliability. If it met this threshold and was otherwise admissible according to the four criteria for expert evidence set out by Sopinka J. in Mohan, supra, then it was up to the jury to determine its ultimate validity and reliability. Case-law is very clear that the voir

dire should not be seen as usurping the role of the jury as final arbiters of the merit of proposed evidence.

Campbell J. gave oral reasons for admitting the DNA testing evidence. He indicated that he intended to give further reasons at a later date but they were not forthcoming. However, his oral reasons set out eight points which he addressed:

1. The qualifications of Ms. Newall, a Crown Forensic Scientist, to testify about various aspects of the technical aspects of DNA profiling and the declaration of a match.
2. The ability of Ms. Newall trained in biology to give expert evidence about the interpretation of human population statistics.
3. Whether the match criteria relied upon by the Ontario Centre of Forensic Sciences is reasonably reliable and, therefore, admissible.
4. Whether the method used by Ms. Newall in the CFS to calculate numerically the statistical rarity of a match in the general population is reasonably reliable and, therefore, admissible.
5. Whether the probative value of those numbers is outweighed by its potential for unfair prejudice.
6. Whether the CFS database yields reasonably reliable results having regard to the racial or ethnic origin of this accused.
7. Does the challenge to the continuity of forensic samples go to the weight or admissibility of the tests yielded from the samples.
8. Is it inadmissible hearsay for Ms. Newall to refer to the results of the FBI computer process applied to the autorads, or to testify about her reliance on Professor Weir's equilibrium test of the CFS database or as to her

conversation with Professor Jefferies.

Campbell J. referred to Ms. Newall's extensive curriculum vitae and to the fact that she had been qualified in other court cases in Ontario. He had no doubt as to her qualifications. He need only have been satisfied that she possessed sufficient skill, knowledge or experience concerning the subject-matter of her expertise and that the proffered opinion would likely aid the trier of fact in reaching a just determination. This condition is satisfied if the witness possesses special knowledge "going beyond that of the trier of fact" (Mohan, supra, at p. 25 S.C.R., p. 255 C.R.). Ms. Newall's curriculum vitae amply supports this finding. This satisfies items (1) and (2) above of the issues identified by the trial judge.

Campbell J. dealt with the other grounds on the basis generally that the attacks on Ms. Newall's evidence went largely to weight and interpretation and not admissibility. With this I agree. Having accepted her credentials and defence concessions as to the validity of the technology she was attempting to apply, the extent to which her evidence was subject to criticism was not a matter for the trial judge to consider with respect to admissibility. He had to hear her testimony in order to scrutinize it, but his scrutiny was limited to satisfying himself that it was sufficiently reliable that it should be received, not that he was satisfied beyond a reasonable doubt, or indeed to a lesser standard of proof, that her conclusions were sound and could be acted upon as proven. Such complaints related to weight and were properly a matter for the jury. This point recalls my distinction, introduced earlier in this judgment, between scrutiny of the methodology and the assessment of the application of a methodology. Whether her conclusions were reasonable on the facts was entirely a matter for the jury to decide as triers of fact.

I do propose to deal with items (5) and (8) since they were the subject of considerable discussion in this court. As to (8), the alleged inadmissible hearsay, I am satisfied that the opinion evidence of Ms. Newall was admissible notwithstanding that the Crown did not call as witnesses the

technicians and other persons upon whose research and reports she relied. Specifically, I do not accept the appellant's submission that there was an obligation on the Crown to tender evidence that did not rely on hearsay statements and reports within the scope of her expertise: a burden not carried by an expert when testifying before the jury. To the extent that her opinions could be criticized for reliance upon other persons' work, it was only the sufficiency of her testimony that was engaged, not its admissibility. This matter was dealt with by Wilson J., writing for the majority of the court in *R. v. Lavallee*, [1990] 1 S.C.R. 852 at p. 896, 55 C.C.C. (3d) 97 at p. 130:

In my view, as long as there is some admissible evidence to establish the foundation for the expert's opinion, the trial judge cannot subsequently instruct the jury to completely ignore the testimony. The judge must, of course, warn the jury that the more the expert relies on facts not proved in evidence the less weight the jury may attribute to the opinion.

There is a further analysis of the admissibility of opinion evidence in *Lavallee* by Sopinka J. He picks up on Wilson J.'s summary of the Supreme Court's decision in *R. v. Abbey*, *supra*, and explains the philosophical basis for distinguishing between classes of hearsay that are relied upon by expert witnesses. His reproduction of Wilson J.'s summary is at p. 898 S.C.R., p. 131 C.C.C.:

1. An expert opinion is admissible if relevant, even if it is based on second-hand evidence.
2. This second-hand evidence (hearsay) is admissible to show the information on which the expert opinion is based, not as evidence going to the existence of the facts on which the opinion is based.
3. Where the psychiatric evidence is comprised of hearsay evidence, the problem is the weight to be attributed to the opinion.

4. Before any weight can be given to an expert's opinion, the facts upon which the opinion is based must be found to exist.

As Sopinka J. points out, the combined effect of numbers 1, 3 and 4 above is that an expert opinion relevant in the abstract but based entirely upon unproven hearsay (as in the case of statements from the accused) is admissible but entitled to no weight. He goes on to say at p. 899 S.C.R., p. 132 C.C.C.:

The resolution of the contradiction inherent in *Abbey*, and the answer to the criticism *Abbey* has drawn, is to be found in the practical distinction between evidence that an expert obtains and acts upon within the scope of his or her expertise (as in *City of St. John*) and evidence that an expert obtains from a party to litigation touching a matter directly in issue (as in *Abbey*).

(Emphasis added)

I had occasion to deal with opinion evidence in an earlier appeal, *R. v. Scardino* (1991), 6 C.R. (4th) 146, 46 O.A.C. 209 (C.A.), and am taking the liberty of repeating what I said at p. 153:

In my view, there is no error in the trial judge's charge. Indeed, it is clear that she patterned her instruction on *Kirkby* [*R. v. Kirkby*, 47 C.R. (3d) 97 (Ont. C.A.)], which in turn relied upon *Abbey*. If any problem arose from *Abbey*, it was a tendency on the part of some judges to rule that before an expert's opinion was admissible in evidence, all the facts upon which the opinion was based must be proved in evidence. *Kirkby*, however, makes it clear that the burden is not that onerous. An expert's opinion is admissible in evidence, notwithstanding the absence of proof in some areas relied upon by the expert. However, the weight to be given to the opinion in such cases is diminished, sometimes to the point where the opinion can be given no weight at all. In my opinion, this view is supported by the recent decision of the Supreme Court of Canada in *Lavallee*, *supra*.

As to Campbell J.'s item (5), the question of whether the probative value of the numbers reflecting the statistical rarity of a match in the general population is outweighed by its potential for unfair prejudice, I would comment firstly that this is an unusual case to raise that argument. While only the Crown called evidence on the DNA profile on the voir dire, competing witnesses testified before the jury: Pamela Newall and Dr. Wayne for the Crown; Dr. Libby, Dr. Mueller and Mr. Coonan for the defence. Accordingly, the jury was not confronted by a single monolithic expression of match significance. The different experts who testified about statistics expressed differing views of the significance of the matches. The experts' views differed largely in how conservative one must be in order to ensure that the significance of a match is not overstated.

Additionally, the DNA evidence was not restricted to one sample. There were samples of blood and semen taken outside the boiler room where the victim was found and further samples from the victim's leotards. The jury was given frequency numbers that ranged from one in 1,500 to one in 1.8 million. The appellant concedes the admissibility of qualitative expressions of match significance (such as "rare" or "common") without the specifics afforded by statistics where DNA evidence is admitted showing a match between the DNA found on the crime scene and the DNA of a suspect, counsel for the appellant objects simply to the admission of the numbers themselves. In this case, it would be difficult to translate the figures the experts were prepared to use into neutral language, but all that aside, why would the defence want to do so? The fact that there are competing figures which differ so radically should be before the jury for its assessment. The range of numeric frequency determined by the various experts was fertile ground for cross-examination. This is a classic case for the application of the language of Dickson C.J.C. in *R. v. Corbett*, [1988] 1 S.C.R. 670 at p. 692, 41 C.C.C. (3d) 385 at p. 400:

The very strength of the jury is that the ultimate issue of guilt or innocence is determined by a group of ordinary citizens who are not legal specialists and who bring to the legal process a healthy measure of common sense. The jury is,

of course, bound to follow the law as it is explained by the trial judge. Jury directions are often long and difficult, but the experience of trial judges is that juries do perform their duty according to the law. We should regard with grave suspicion arguments which assert that depriving the jury of all relevant information is preferable to giving them everything, with a careful explanation as to any limitations on the use to which they may put that information.

I do not believe that there should be an absolute prohibition against the introduction of specific match figures. The appellant correctly notes that the case-law reflects conflicting conclusions as to the admissibility of DNA probability statistics. It was justifiable to admit the probability statistics in this case, and it might be in others. I would leave the matter to the discretion of the trial judge in the particular case.

As I indicated earlier, Campbell J. did not have the benefit of the reasons of Sopinka J. in *Mohan*, supra, but his ruling on the admissibility of the DNA evidence was in the face of objections to its validity and reliability which the trial judge acknowledged in setting out the issues on the voir dire. Accordingly, it is clear that he was satisfied at least that it was reliable enough that he could not preclude its admission into evidence. Indeed, there is much force to the argument of the Crown that with the concessions as to the reliability of the DNA technology in the abstract, novelty was not in issue, and the dispute as to whether it was properly applied by the Crown's experts in the particular case was exclusively within the province of the jury. I do not have to go that far, but if my interpretation of what was required of the trial judge by *Mohan* is correct, it is apparent that the argument in this appeal was somewhat academic given the thoroughness with which the reliability of the DNA evidence was canvassed prior to its admission into evidence.

As to the issue of the appropriate standard of proof on the voir dire, the trial judge, to the extent that he made findings of fact, applied the standard of a balance of probabilities. This is apparent from his reliance upon the trial decision in

R. v. Johnston, supra. He was correct in doing so. The issue of reliability respecting novel scientific theory or technique relates strictly to a question of the admissibility of evidence where proof on a balance of probabilities is an acceptable standard: see R. v. B. (K.G.), [1993] 1 S.C.R. 740, 79 C.C.C. (3d) 257. This is not an inculpatory statement made by an accused to a person in authority ([CF2]B. (K.G.), supra, at p. 800 S.C.R., p. 297 C.C.C.) nor is it the establishment by the Crown of "facts which trigger a presumption with respect to a vital issue relating to guilt or innocence": R. v. Egger, [1993] 2 S.C.R. 451 at p. 475, 21 C.R. (4th) 186 at p. 202.

The same standard, balance of probabilities, applies to the qualification of an expert witness even where the science is novel. The trial judge need not be satisfied that one expert witness is qualified on a balance of probabilities and as to another beyond a reasonable doubt. Ms. Newall's qualifications as an authority on DNA testing were not questioned per se. To the extent that they were, Campbell J. left no doubt that he accepted her as eminently qualified in the field.

It must be remembered that we are dealing with the admissibility of the opinion of an expert witness with respect to one piece of circumstantial evidence relevant to the identity of the perpetrator of a crime. I can think of no justification for imposing a special burden of proof upon the Crown with respect to these DNA experts. This after all is still identification evidence. Scientific methods of identification, including analysis of bodily fluids such as blood, semen, saliva, hair, as well as fingerprints, footprints, dental impressions, and striations on bullets, all depend upon the ability to match samples, one to another. DNA profiling differs from this earlier technology only in its increased power to discriminate between individuals.

I think that the fundamental error propounded by the counsel for the appellant is his submission that Mohan has introduced, as a precondition to admissibility, a new standard of proof in the scrutiny of opinion evidence that relates to a novel scientific theory or technique. If the submissions of the appellant are to be accepted, there would be a burden upon the

Crown to satisfy the trial judge beyond a reasonable doubt that its proffered evidence was reliable. Additionally, in giving testimony on the voir dire, the expert could not rely upon opinions and tests supplied by others that are within the field of his or her expertise, a restriction that is contrary to all established authority.

The combination of these two changes would award to DNA evidence a special status not accorded to the opinions of experts generally in criminal trials. Counsel for the appellant submits that this result is justified because DNA evidence is so overwhelmingly compelling that it is dispositive of the guilt or innocence of the appellant. He submits that DNA evidence justifies introducing as a matter of procedure an hermetically sealed trial within a trial wherein reliability must always be separately assessed to the criminal standard of proof beyond a reasonable doubt as a precondition to admissibility. He relies on the words of Sopinka J. in *Mohan*, supra, where he states at p. 25 S.C.R., p. 255 C.R.: "The closer the evidence approaches an opinion on an ultimate issue, the stricter the application of this principle".

The notion that a voir dire is held to pass on the sufficiency of evidence has been rejected by this court in wire tap cases in *R. v. Parsons* (1977), 17 O.R. (2d) 465 at pp. 468-69, 37 C.C.C. (2d) 497 at pp. 500-01. There Dubin J.A. for a panel of five judges made the point that the suggestion that the trial judge must hear all the evidence led with respect to each recorded conversation and weigh that evidence "would be to misconceive the purpose of a voir dire, and confuse the respective functions of a Judge and jury in a criminal trial . . . A voir dire is not held to pass on the sufficiency of the evidence, but only to determine questions of admissibility".

This point is emphasized by what was said recently by the Supreme Court of Canada in *R. v. U. (F.J.)*, [1995] 3 S.C.R. 764, 101 C.C.C. (3d) 97, on the content of and standard of proof in a voir dire as it applied to a statement made to the police by a complainant in a sexual assault case which she later retracted at trial. At pp. 793-95 S.C.R., pp. 119-20 C.C.C., Lamer C.J.C. repeated what he said on this subject in

R. v. B. (K.G.), *supra*, that the voir dire is to be held so that the trial judge can be satisfied on a balance of probabilities that the initial statement was not the product of coercion in any form. He then said at p. 795 S.C.R., p. 120 C.C.C.:

The trial judge at this stage is not making a final determination about the ultimate reliability and credibility of the statement. The trial judge need not be satisfied that the prior statement is true and should be believed in preference to the witness's current testimony.

If the trial judge determines that the statement meets the threshold reliability criterion and is thus substantively admissible, he or she must direct the trier of fact to follow a two-step process in evaluating the evidence.

The appellant relies upon the accepted onus on the Crown in determining the admissibility of confessions. The onus is described by Sopinka, Lederman and Bryant, *The Law of Evidence in Canada* (Toronto: Butterworths, 1992) at p. 359 as follows:

The present law in Canada and the U.K. is that the prosecution must prove voluntariness beyond a reasonable doubt. This view is based on the reasoning that since a confession is potentially determinative of the issue of guilt or innocence, the criminal standard of proof should be maintained.

The appellant also relies on R. v. Egger, *supra*, a breathalyzer case, where the Supreme Court of Canada held that the standard of proof for the service on the accused of a Certificate of Analysis and a Certificate of Qualified Technician within the times proscribed by the Criminal Code, R.S.C. 1985, c. C-46, was to the criminal standard of beyond a reasonable doubt before the Crown could rely upon the presumption in the certificate as to the blood-alcohol content of the accused's blood. However, this does not engage a question of the admissibility of evidence. As Sopinka J. stated at pp. 474-75 S.C.R., p. 202 C.C.C.:

The issue here is very different from a question of admissibility of evidence. The effect of satisfying the burden of proving preliminary facts to the admissibility of evidence is only that the evidence is admitted: it determines neither the weight of the evidence nor the guilt of the accused. This occurs in the next step in the process during which the Crown must establish its legal burden. When admission of the evidence may itself have a conclusive effect with respect to guilt, the criminal standard is applied. This accounts for the application of this standard with respect to the admission of confessions (see *Ward v. R.*, [1979] 2 S.C.R. 30 at p. 40, per Spence J., for the Court, and *R. v. Rothman*, [1981] 1 S.C.R. 640, at pp. 670, 674-675, per Martland J., for the majority, and at p. 696, per Lamer J. (as he then was), concurring).

To return to the quotation from Mohan as to the need for stricter scrutiny the closer the evidence gets to "an ultimate issue", the ultimate issue with respect to DNA profiling is not guilt or innocence as with a confession. It is but one piece of circumstantial evidence which taken alone may prove very little. In the case in appeal, much of the hair, fibre, blood and DNA evidence which puts the appellant at the scene of the crime is, on its face, relatively neutral given that the appellant was one of the janitors for this apartment building and had a prior innocent association with the victim. It gains greater significance, however, when considered in the light of the appellant's progression of statements as to his limited access to the crime scene as evidence of consciousness of guilty and the statements of Crown witnesses, among other factors present at the trial. The defence scoffs that without this and the other forensic evidence there would not be a case for the appellant to meet. That may or may not be the case in the matter that is before us, but I can think of other cases where DNA profiling and other forensic evidence was in the last analysis superfluous. The fact that the DNA evidence may be more important to the strength of the Crown's case in this instance is not a reason for raising the level of scrutiny on the voir dire as to admissibility. The defence should properly concentrate, as it did here, in attempting to cast doubt upon its accuracy in the presence of the jury.

The real concern of the defence arises out of the accuracy of the DNA testing. It is the high degree of probability of the matches that creates the "mystic infallibility" of the DNA evidence in the eyes of the jury. However, the concern as to accuracy is to a large degree offset by the availability of the DNA samples for independent sampling at the instance of the defence. In the case on appeal, the samples relied upon by the Crown were available for further testing. In my review of the record of the voir dire, I did not find any suggestion by defence counsel that the quality of the DNA samples that were tested prohibited the samples from being examined or tested further. In fact, during the trial judge's ruling, the admissibility of the DNA evidence was considered in the context of a number of separate headings. Under the heading, "Depletion of Sample Material", the trial judge ruled as follows:

The defence does not proceed with any potential Charter motion relating to unavailability through depletion of sample material for further testing.

In the case before the court, the defence had its own experts who testified before the jury. They were critical of various technical matters leading up to the creation of the autorads, the misreading of the bands, the lack of control, the relative newness of the CFS's experience in DNA typing, the accuracy of the match probabilities and other matters. I think the language of Moldaver J. in *R. v. Melaragni*, supra, is applicable. He said at p. 354:

I am equally convinced that the jury will not be overwhelmed by the "mystic infallibility" of the evidence. . . . I have no doubt that the jury will carefully consider cross-examination designed to weaken or destroy the worth of the proposed evidence. I have every confidence that the jury will pay close attention to an opposing expert, and I am equally confident that the jury will follow legal instruction regarding the worth of expert evidence in general and this evidence in particular.

Campbell J. dealt explicitly in his charge with the concern

that the jury might be overwhelmed by DNA profiling evidence:

The DNA tests in this case were conducted soon after the Centre of Forensic Labs opened itself for DNA case work. And there is evidence that challenges the conduct of the DNA tests and evidence that challenges the results. These are reasons for you to take a good close look at the DNA evidence, yourself, all the evidence you've heard from the Crown and the defence, and scrutinize it to see if you consider it reliable as a piece of circumstantial evidence. You have obviously followed it closely, spent a lot of time in court looking at the autorad projections, various aspects of the bands and their measurement and interpretation. You've seen those from both sides. I don't intend to repeat all of that. You have part of it, or small parts of it, anyway, in some of the material in front of you. You followed it very closely. I am confident you will use your common sense, you won't be overwhelmed by any aura of scientific authority advanced by any of the DNA witnesses. The assessment of the evidence really does boil down to a common sense assessment of the evidence, of the various opinions that you have heard, your assessment.

A further point must be made on the ultimate issue. This DNA evidence does not tend to establish conclusively the identity of the perpetrator of the crime. In a given case, it can only eliminate conclusively a suspect. DNA profiles are designed to determine if the appellant's genetic makeup is consistent with the genetic makeup of the samples taken at the scene of the crime. In order to indicate the degree of consistency, the expert witness will normally provide both quantitative and qualitative statements directed at the probability of randomly matching an individual in the population to the DNA samples present at a crime scene. The problem with using qualitative modifiers such as rare, unlikely and remote is that they are awkward and fail to convey the potency of a match. For this reason, the scientific community seems to prefer to use specific figures, as in this case. On the other hand, the underlying concern of the defence is that the jury will be permitted to fall into what is referred to as "the prosecutor's fallacy": equating the probability of a random match with the

probability of the appellant's innocence. In other words, the concern is that the jury will convert the statistics into something approaching the ultimate issue. The conclusion that may be drawn from probability statistics is that it is rare or common to find this pattern among known DNA samples; one cannot make the leap to conclude that as a result of a match frequency the DNA found on a scene is that of a particular suspect.

However, counsel for the appellant acknowledges that there was no prosecutorial fallacy in this case and the instruction to the jury makes plain that the procedures employed in the DNA profiles simply generated statistics. The trial judge was very clear in his instructions to the jury as to how it should use them. The jury charge includes explicit instructions explaining both how the statistical evidence should not be used (e.g., only one in 3.7 million people have this profile) and how it should be used (i.e., to provide an understanding of the degree or rarity of the profile). Moreover, the trial judge told the jury that the DNA evidence was just one piece of circumstantial evidence in this case. On this record, there is no basis for an inference that the jury would have used the statistics as a predictor of the likelihood of guilt.

As noted above, my opinion is that the process of arriving at the point where a trial judge is satisfied as to the threshold of reliability for novel scientific theory or technique should be a flexible one. While I am cognizant that the standard of scrutiny varies depending upon how close the evidence approaches an opinion on an ultimate issue, I think that, once again, this evaluation is particular to the facts of the case. DNA matches do not decide an ultimate issue, they are a significant piece of circumstantial evidence that in this case, taken with the other evidence, support the theory of the Crown that the appellant was at the scene of the crime on the date in question and that he sexually assaulted the victim.

The last point relating to DNA evidence considers the appellant's objection to the trial judge's instruction on the DNA evidence. I believe that I have reviewed the subject indirectly, because the most significant objections related to reassertions of the arguments that the basis for the opinion

evidence of the Crown had not been established. I have considered the instruction in its entirety and have reviewed the arguments of the appellant's counsel. In my view there is no substance to these objections. The instruction was in accordance with the judgments in *Abbey* and *Lavallee*, *supra*.

I will deal with one specific objection. In the charge to the jury, the trial judge's lengthy instructions concerning DNA evidence contained the following passage:

I turn to DNA as circumstantial evidence. Remember it's only one form of circumstantial evidence. You do not apply the standard of proof beyond a reasonable doubt to the individual pieces of circumstantial evidence upon which the Crown relies. I say it again. I will say it at least one more time: Don't get bedazzled or unduly swayed by some of the large numbers used in the DNA evidence. Remember, also, the burden of proof is always on the Crown, the burden never shifts, there is never any obligation on the defence to prove anything.

At trial, defence counsel objected to this aspect of the charge, suggesting that Campbell J. ought to have directed the jury that "if the experts are in a quandary and the jury is in a quandary then they are simply to set it aside". In my opinion, the trial judge was correct in declining to instruct the jury in the manner suggested by defence counsel. The appellant submits that DNA evidence should be treated differently from other forms of circumstantial evidence and that the jury ought to be instructed that the reasonable doubt standard applies to this individual piece of evidence. This approach to expert evidence, and DNA evidence in particular, is inconsistent with the judgment of the Supreme Court of Canada in *R. v. Morin*, [1988] 2 S.C.R. 345, 44 C.C.C. (3d) 193. In that case, Sopinka J. held at pp. 360-62 S.C.R., pp. 210-11 C.C.C.:

The argument in favour of a two-stage application of the criminal standard . . . is wrong in principle and unworkable in practice. In principle, it is wrong because the function of a standard of proof is not the weighing of individual

items of evidence but the determination of ultimate issues. Furthermore, it would require the individual member of the jury to rely on the same facts in order to establish guilt. The law is clear that the members of the jury can arrive at their verdict by different routes and need not rely on the same facts. Indeed, the jurors need not agree on any single fact except the ultimate conclusion.

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I conclude from the foregoing that the facts are for the jury to determine subject to an instruction by the trial judge as to the law. . . . [T]he law lays down only one basic requirement: during the process of deliberation the jury or other trier of fact must consider the evidence as a whole and determine whether guilt is established by the prosecution beyond a reasonable doubt. This of necessity requires that each element of the offence or issue be proved beyond a reasonable doubt. Beyond this injunction it is for the trier of fact to determine how to proceed. To intrude in this area is . . . an intrusion into the province of the jury.

Summary respecting admissibility of novel science

I do not believe that the judgment of the Supreme Court of Canada in *Mohan* intended to introduce a new format for the conduct of a voir dire for the scrutiny of novel scientific theory or techniques in order to establish a threshold of reliability. The screening process is directed to the issue of the admissibility of the novel expert testimony, not the determination of its ultimate reliability. I can see nothing in recent authority which indicates a change in the traditional role of the trial judge in assessing the reliability of proffered evidence before ruling on its admissibility. He or she is not the trier of fact and should not invade the province of the jury by making findings of fact on the ultimate issues. The trial judge's function is limited to an overview of the evidence proffered in order to be satisfied that it reflects a scientific theory or technique that has either gained acceptance in the scientific community, or if not accepted, is considered otherwise reliable in accordance with the methodology validating it. The trial judge

will be required to hear sufficient evidence to determine reliability as a preliminary matter. Moreover, the trial judge must not pass judgment on the particular application of the methodology by the expert. This is a question of weight to be determined by the jury. The trial judge must restrict his inquiry to determining whether the proposed novel scientific technique or theory has a foundation in science, as determined. The nature and scope of the evidence necessary for the trial judge to reach the threshold determination will vary according to the type of evidence proffered and the concessions made by counsel. As a result, it would be unwise to define the threshold test of reliability with the precision advanced by the scientific community. Rather, the threshold test of reliability must remain capable of adaptation to changing circumstances and realities. Reliability is best determined under the scrutiny of the trial judge as guided by the demands and particularities of the case. Simply stated, the threshold test of reliability is met when the trial judge, having reviewed certain evidence presented by counsel, feels that the novel scientific technique or theory is sufficiently reliable to be put to the jury for its review.

At the conclusion of the evidence, the trial judge in his instruction should advise the jury in the normal way as to the limits of the expert evidence and the use to which it can be put. Additionally, in the case of DNA evidence, he or she would be well advised to instruct the jury not to be overwhelmed by the aura of scientific infallibility associated with scientific evidence. The trial judge should tell them to use their common sense in their assessment of all of the evidence on the DNA issue and determine if it is reliable and valid as a piece of circumstantial evidence.

(2) The admissibility of a prior consistent statement given to the police by the appellant

On December 3, 1990, shortly after noon, the appellant was arrested for the first degree murder of Andrea Atkinson. At 12:52 p.m., he was interviewed by the arresting officers, Detectives Gauthier and McNamara and made an exculpatory statement. The appellant had previously provided four statements to the police, two on October 23, 1990, one on

October 25, 1990, and another on October 30, 1990. The Crown introduced all of the appellant's statements at trial except for the December 3 statement.

The defence wanted to introduce the appellant's exculpatory statement of December 3, 1990 as part of its case for essentially four reasons:

- (i) to show the jury that the appellant provided an exculpatory statement when he was arrested and at a time when the police first confronted him with all the forensic and other evidence that incriminated him;
- (ii) because the appellant offered for the first time an explanation as to why his hairs and fibres were found on the sixth floor, an explanation that was consistent with his testimony at trial; and,
- (iii) as a basis to cross-examine the detectives concerning the manner in which they cross-examined the appellant during the statement that he gave after his arrest in the context of their overall investigation of the appellant;
- (iv) to rebut the Crown's allegation of recent fabrication.

The defence position at trial was that the statement was admissible as a continuation of the appellant's earlier statements; as part of the *res gestae*; in order to rebut the prosecution's anticipated allegation of recent fabrication; as a prior consistent statement in accordance with the decision in *R. v. Small*, unreported, September 11, 1991 (Ont. Gen. Div.), and in order for the appellant to make full answer and defence.

The Crown points out that the appellant's December 3, 1990 statement to Detective Sergeant Gauthier and Detective McNamara was made seven weeks after Andrea was murdered and forty days after her body was discovered. The appellant had made his initial statements to the police and first provided bodily samples more than a month after he was advised that he was a suspect in a murder investigation and "may be charged with murder". The statement in issue was made more than two weeks

after a search warrant was executed at the Terceira residence and about 30 minutes after his arrest.

The trial judge held that the statement was inadmissible. He also refused to allow the defence to inform the jury of the fact that the appellant was questioned when he was arrested, and that he provided a statement. He ruled that:

The defence proffered by the appellant was the same in each of his statements, namely "I didn't do it". His alibi was put forward from the outset. The appellant was confronted with the presence of his hair and semen in the course of his October 30, 1990 statement. Although the police had more to confront the appellant with on December 3, his complete denial of guilt was identical on each occasion and there is no new explanation by the accused in December 3 that was omitted from the earlier statements.

Campbell J. stated that he was bound by the decision of the Ontario Court of Appeal in *R. v. Campbell* (1978), 17 O.R. (2d) 673, 38 C.C.C. (2d) 6, the principles of which had been affirmed by the Supreme Court of Canada in *R. v. Beland*, *supra*, at pp. 409-12 S.C.R., pp. 489-91 C.C.C. and therefore declined to follow *R. v. Small*, *supra*. He also expressed a willingness to revisit his ruling in the event that the Crown raised the issue of recent fabrication, either expressly or impliedly.

In *Campbell*, *supra*, this court held that defence counsel was not entitled to elicit the accused's prior exculpatory statements during cross-examination of the police and customs officials to whom they were made, and equally that he should have been precluded from narrating his prior statements during his own testimony. The position in Canada has traditionally been that prior consistent statements are irrelevant and superfluous. Although the appellant's counsel acknowledges that statements by an accused have generally been ruled inadmissible in Canada, he submits that certain of the authorities relied upon by Martin J.A. of this court in *Campbell*, *supra*, do not support the ratio of the decision. He submits that Martin J.A. ought to have adopted the English exception to the rule against prior consistent statements. He refers to a number of these

authorities starting with *R. v. Storey* (1968), 52 Cr. App. R. 334, wherein the English Court of Appeal dealt with a voluntary exculpatory statement by the accused that had been introduced by the Crown as part of its case. The court held that such a statement was admissible simply to show the reaction of the accused when he or she was "first taxed with incriminating facts" but is not admissible for the truth of its contents. Subsequent cases in England have expanded the *Storey* exception to permit exculpatory statements by an accused to show reaction or attitude not restricted to statements made by the accused when the accused is "first taxed" with "incriminating facts" during the "first encounter with the police".

In my view, *Storey* is distinguishable on its facts from the case in appeal. In any event I do not think that this is an appropriate case to re-examine the basic thinking behind *Campbell* as we were invited to do by counsel for the appellant. The ruling by the trial judge that the statement of December 3, 1990 is but one of a number of exculpatory statements made after the appellant was well aware of his status as a suspect is fully supported by the record. Moreover, as noted by the trial judge, the December 3, 1990 statement was so similar in substance to earlier statements that no new or helpful information would be provided by admission of the December 3, 1990 statement during the trial proper. The time that had elapsed from the first statement to the one in issue was too long to support a suggestion that the five statements were part of a continuum. The invitation of the trial judge to revisit his ruling in the event that the Crown raised the issue of recent fabrication was never accepted by the defence despite the position now advanced that the Crown cross-examined the appellant on all five statements made by the appellant. Further, no objection was taken by the defence to this line of questioning nor did the defence ask to re-examine on the fifth statement. I would not give effect to this objection.

(3) The admissibility of hypnotically refreshed testimony

The appellant's evidence as to his whereabouts at the time of Andrea's disappearance was supported by the testimony of his mother and grandmother. They testified that the appellant was

home shortly after 11:00 a.m. on the day Andrea went missing. David Belanger, who lived in the same apartment building, gave evidence that he saw Andrea alive at 12:00 p.m. that day. To rebut the appellant's alibi, the Crown called two tenants who claimed to have seen the appellant working at the apartment building around 1:30 p.m.

One of these witnesses was Corinna MacNaughton. MacNaughton gave statements to the police on October 28, 1990 and November 11, 1990 which contained a number of aspects which suggested that she was recalling the events of another day. In light of the importance of MacNaughton's evidence, the police arranged for her to see Dr. Matheson, a psychologist, in order for her to be hypnotized as a way of improving her memory. Before her visit with Matheson, the police advised her of "errors" in her memory of the events of the day that Andrea disappeared. Following her hypnosis, MacNaughton gave a third statement to the police which was consistent with the testimony she provided at trial. At trial, she testified that the hypnosis helped her remember the details of that day:

Just certain details: Cleaners that I thought I had seen that day that I did, in fact, see that day -- that I didn't see that day, it was just a different day that I had seen it.

The defence objected to the admissibility of what counsel for the appellant now describes as MacNaughton's hypnosis induced testimony. Counsel argued that her evidence was now unreliable having been tainted by the hypnosis and by the police telling her that certain events she was recalling were impossible. A voir dire was held. Dr. Matheson was called by the Crown as well as Constable Johnston who had interviewed MacNaughton. Dr. Matheson explained the process of hypnosis.

It is evident that while defence counsel objected to the testimony of MacNaughton, it was on the basis that her evidence was tainted, not by what took place during the hypnosis, but by information that had been provided to her by the police. At no time did trial counsel challenge the validity of hypnosis as a reliable scientific technique nor did he challenge Dr. Matheson's expertise in the area of forensic hypnosis.

The primary focus of defence counsel, in his cross-examination and submissions during the voir dire, was on the tainting of MacNaughton's evidence by the police. Constable Johnston acknowledged that, during an interview prior to the hypnosis, the police drew to Ms. MacNaughton's attention certain inaccuracies in her recollection and "corrected her". Dr. Matheson testified that Ms. MacNaughton's memory "might" have been altered by the police correction, but that any such alteration was independent of what had occurred in the hypnosis session. During his submissions, defence counsel relied upon tainting of the witness's memory by the police, but did not suggest that her recollection had been affected by the hypnosis itself.

Defence counsel on the voir dire declined to take issue with the reliability of hypnosis as a means of refreshing memory. To the extent that there was any evidence provided during the voir dire which spoke to the issue of reliability of hypnosis, the evidence before the trial judge during the voir dire on the issue was that hypnosis was accepted within the medical profession and by professional bodies as a means of refreshing memory. Furthermore, having regard to defence counsel's attack on the reliability of DNA testing as a scientific technique, it seems reasonable to infer that he made a strategic decision to deal with MacNaughton's testimony as an issue of witness tainting. The court does not readily permit an appellant to reverse such a tactical decision on appeal.

During the trial proper, defence counsel conducted an effective cross-examination of Corinna MacNaughton in which he repeatedly pointed out inconsistencies between her statements to the police and her testimony at trial, impeached her credibility through use of the transcript of her hypnosis session with Dr. Matheson, and highlighted deficiencies in her recollection which were unaffected by hypnosis. Further, during the testimony of MacNaughton and Constable Johnston, defence counsel brought out the areas in which the police had "corrected" her recollection.

The complaint is further made that while Dr. Matheson was

available to be called by either party when the MacNaughton testimony was presented to the jury, there was an obligation upon the Crown to call the psychologist. The Crown at trial took the position that since the defence focus had been on police tainting during the voir dire, rather than any tainting specific to the hypnotic interview, it chose not to call Dr. Matheson as a witness at trial. The transcript is equivocal as to whether defence counsel acquiesced in this position. However, it is clear that defence counsel did not object to the Crown's decision, did not ask the learned trial judge to call Dr. Matheson as a witness, did not choose to call the doctor as a defence witness, and chose not to call a defence expert on hypnosis.

I have some difficulty in understanding what the argument on appeal has to do with the admissibility of Corinna MacNaughton's evidence. Even accepting that it was "hypnotically induced", this has nothing to do with whether she should be permitted to testify before the jury, particularly in light of the fact that the reliability of hypnosis was not questioned by defence counsel. As a result, in the present case, the effect of hypnosis on the memory of the witness was properly a matter going to weight to be decided by the trier of fact. The Crown put all its cards on the table on the voir dire and the defence determined that the witness was vulnerable to the suggestion that her evidence was tainted. Defence counsel cross-examined her vigorously on this theme before the jury and asked for and received from the trial judge an instruction to the jury as to the special caution that should be exercised in assessing her evidence due to the fact that she had been hypnotized. I would not give effect to this ground of appeal.

(4) The propriety of permitting the Crown to call Dr. Humphries in reply

While the experts agreed that there were substantial indicia of asphyxia as the cause of Andrea's death, the defence disputed the condition that may have triggered the asphyxia. The Crown position was that the appellant had caused the asphyxia by placing his hands over Andrea's mouth. The defence

position was that Andrea had suffered an epileptic seizure triggered by the sexual assault and a pre-existing head injury. According to the defence, it may have been the seizure which caused the asphyxia. Consequently, assuming the jury was satisfied that the appellant was the assailant, there was evidence which if believed could lead to a verdict of manslaughter as opposed to murder.

At trial, the Crown called Dr. Noel McAuliffe, a pathologist from the Forensic Pathology Branch of the Chief Coroner's Office, who had performed the autopsy on Andrea. He testified that he had been informed, at the preliminary inquiry, of a head injury that Andrea had suffered six weeks prior to her death. It was his opinion that the injury he observed at the autopsy was not related to this earlier injury. The skull injury he observed at the autopsy was a post-mortem development caused by a build-up of gas pressures. He was cross-examined by the appellant's counsel about the link between head injuries, epileptic seizures and asphyxia:

Q. And that type of bump or injury pre-existing, one of the side effects, I am suggesting to you, could very well have been the onsetting of a seizure, an epileptic seizure?

A. Certainly head injuries have the potential to initiate epilepsy, that's right.

Q. And people who suffer seizures may, in fact, never exhibit visible brain abnormality or that sort of thing, is that correct?

A. That would be true in the majority of cases, yes.

Q. And when one has a seizure which results in death that death would be by way of asphyxia?

A. Yes.

The Crown re-examined Dr. McAuliffe regarding the defence position. The Crown elicited from Dr. McAuliffe that Andrea had

no prior head injury of sufficient magnitude to cause a seizure six weeks later:

The emergency chart would suggest that the injury was interpreted as nothing above absolutely trivial. An examination was made and noted and a code for billing OHIP was inserted. It would seem to be an absolutely trivial matter. . . . If there was a report of a fracture on an X-ray taken, at the time, I would think it would still have absolutely no relationship to the enormous fracture that we now see. I don't -- I don't believe for a moment that this child can be carrying on about her business for the intervening, almost two months with a fracture to the extent that I found at the autopsy.

The Crown called no further witnesses in its own case regarding this issue. The defence called as part of its case Dr. James Ferris, a forensic pathologist from the University of British Columbia. Dr. Ferris challenged the opinion of Dr. McAuliffe. In his opinion, the skull injury noted by McAuliffe at the autopsy resulted from a pre-existing injury to Andrea's head, an injury which could have triggered an epileptic seizure which in turn could have caused the asphyxia.

The Crown sought to call Dr. Robin Humphries, a paediatric neurosurgeon from the Hospital for Sick Children, in reply. The trial judge ruled that he could testify in reply. The trial judge was of the opinion that Dr. Ferris, in his evidence, had introduced, for the first time, the issue of Andrea having suffered a "growing" fracture. In his opinion, Dr. McAuliffe had only been cross-examined by the appellant's counsel concerning the possibility of a "healing" fracture. Campbell J. ruled, in part, as follows:

This theory of an injury which had healed leaving a brain injury underlying the fracture site which later resulted in some sort of spontaneous death through asphyxiating of epilepsy was, in my view, quite different from the evidence of Dr. Ferris. Although Dr. Ferris did on a couple of occasions refer to it as a "healing injury".

The impact of Dr. Ferris's evidence, and its overall impression, was quite different from the picture put in cross-examination. The most forceful part of his evidence was his testimony that it was a particular thing that was never put to Dr. McAuliffe, that is a growing fracture.

Dr. Humphries' evidence in reply was that Andrea's skull injury observed by Dr. McAuliffe was neither a growing or healing fracture. He agreed with Dr. McAuliffe that it was a post-mortem artifact. Dr. Humphries further testified that even if Andrea had suffered a head injury six weeks prior to her death, her death was unrelated to it.

The test for reply evidence was recently articulated in *R. v. Melnychuk*, [1997] 1 S.C.R. 602, 114 C.C.C. (3d) 503, reversing (1996), 104 C.C.C. (3d) 160 (Ont. C.A.) at pp. 172 and 174, where the dissenting judgment of Doherty J.A. of this court was subsequently adopted by Sopinka J., for the majority of the Supreme Court of Canada:

The case law recognizes . . . that a trial judge may receive reply evidence which, while of some relevance to the allegations from the outset, takes on real significance only in light of a position advanced during the case for the defence.

Defence evidence that conflicts with Crown evidence related to an essential issue opens the door to reply evidence only where the Crown could not foresee the need to lead the evidence as part of its case.

I tend to agree with the appellant that the introduction of the distinction between a "healing fracture" as opposed to a "growing fracture" was something of a red herring. The evidence led through Dr. McAuliffe was that Andrea had died as a result of asphyxiation, not any kind of a head injury. Possibly due to the degree to which her body had decomposed, he was unable to determine the mechanism by which she had asphyxiated.

On the other hand, the defence pathologist, Dr. Ferris, would

have classified the cause of death as "undetermined". He did not agree that the signs of asphyxia were so significant that it could be definitively pronounced the cause of death. Even assuming that Andrea did asphyxiate, Dr. Ferris testified that an epileptic seizure, stimulated by what he described as a pre-existing "growing fracture" of the skull, may have caused the asphyxia. Dr. Ferris was unequivocal that Andrea was suffering from a pre-existing "growing fracture" and that this ante-mortem injury may have resulted in a seizure. He theorized that Andrea may have suffered a spontaneous seizure, coincident in time with the sexual assault on her or that the sexual assault could have acted as a catalyst for the possible seizure.

Contrary to the submissions of the appellant, the Crown was not splitting its case. The fact that Dr. McAuliffe had responded to certain hypothetical questions on the premise that there was a pre-existing fracture did not place an onus on the Crown to chase down and eliminate this possibility that the asphyxia had resulted from a fracture which was not evident to Dr. McAuliffe. It was not until the defence pathologist, Dr. Ferris, put this theory forward affirmatively that it took on "real significance" and presented the jury with an alternative cause of death which could have impacted on the verdict.

The appellant relied upon the decision of the Supreme Court of Canada in *R. v. Biddle*, [1995] 1 S.C.R. 761 at pp. 770-79, 36 C.R. (4th) 321 at pp. 328-34, where, in upholding the judgment of this court, it held inter alia that "rebuttal will not be permitted regarding matters which merely confirm or reinforce earlier evidence adduced in the Crown's case which could have been brought before the defence was made". This case has no application. In *Biddle*, the Crown sought to change its case to meet an alibi defence of which it was apprised after it had closed its case. Had the defence made available to the Crown before the close of its case a medical report of Dr. Ferris setting out his opinion and indicated that the defence proposed to call Dr. Ferris if necessary, the trial judge's ruling might well have been different. Here, the Crown was not changing its position, it was maintaining it by reacting to a

defence witness who suggested a different cause of death.

(5) The instruction to the jury on consciousness of guilt

The appellant contends that the instruction on consciousness of guilt was erroneous, as it failed to direct the jury to apply the standard of proof beyond a reasonable doubt to the evidence of the appellant's lies to the police. The recent decision of *R. v. White* (1996), 29 O.R. (3d) 577, 108 C.C.C. (3d) 1 (C.A.), is dispositive of this issue. In *White* this court, sitting as a panel of five specifically to reconsider this issue, reversed its earlier decision of *R. v. Court* (1995), 23 O.R. (3d) 321, 99 C.C.C. (3d) 237 (C.A.), and held that evidence of consciousness of guilt should not be considered in isolation, and should not have the standard of proof beyond a reasonable doubt applied to it separately from the rest of the evidence. In my opinion, the trial judge's charge met the standard set out in *White* and contains no reversible error.

DISPOSITION

Accordingly, for the reasons stated, I would dismiss the appeal.

Appeal dismissed.